Air University
Final Examinations: Spring 2026

Student ID: 231285

AIR UNIVERSITY

Subject: Database Systems
Class: BSCYS -VI
Section: B
Course Code: CYS-130

Time Allowed: 3Hrs
Max Marks: 100
FM's Name: Ms. Irum Aslam
FM's Signature:

INSTRUCTIONS

- Attempt responses on the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on question paper will be considered as use of unfair means.

CLO 2

Question 1 (4 + 4 + 4 + 4 + 4 = 20 Marks)

For each of the following questions: write (a) the name of the ER concept being described and (b) the shape or symbol used to represent it in an ER diagram.

i. A cybersecurity firm stores records of its analysts. Each analyst has an AnalystID, a name, and a city. AnalystID alone is enough to uniquely identify any analyst in the database. A student argues that AnalystName should also be given the same special marking in the ER diagram because names are important to the business — but the database designer disagrees. What ER concept applies to AnalystID here, why is the student wrong about AnalystName, and how is this concept represented in the ER diagram?

ii. One analyst in the firm holds CEH, OSCP, and CISSP certifications all at the same time. Another analyst holds only CEH. The designer wants to store all certifications for each analyst as a single attribute rather than creating a separate table but realizes a plain ellipse does not correctly capture the nature of this attribute. What type of attribute is Certifications in this scenario and how is it drawn in the ER diagram?

iii. The firm logs Incidents raised by analysts. Each incident has an IncidentID, but this number restarts from 1 for every analyst — so analyst A01 and analyst A02 can both have an incident numbered 1. If an analyst is removed from the database, all their incidents are removed as well. An incident record cannot exist on its own without a linked analyst. What type of entity is Incident, what is IncidentID called in ER terminology, and how are both the entity and its relationship to Analyst drawn differently from a normal entity and relationship?

iv. The firm has a rule that every analyst must be assigned to at least one project before their record can be saved in the system. At the same time, a project record can be created and saved in the system before any analyst is assigned to it — for example when a project is newly approved, but staffing has not started yet. What participation type applies to Analyst in this relationship, what participation type applies to Project, and how is each one shown in the ER diagram?

v. Management wants to see TotalHoursLogged displayed on each analyst's profile page. The DBA points out that this value is never entered into or stored directly — it is always recalculated from the sum of hours across all incidents linked to that analyst. Storing it separately would risk it becoming out of date whenever new incidents are added. What type of attribute is TotalHoursLogged and how is it represented in the ER diagram to show that it is not physically stored?

CamScanner

CLO 3

Question 2 **(10 + 5 + 5 + 10 + 10= 40 Marks)**

A cybersecurity firm tracks its analysts and the projects they work on. All data is stored in one flat table ANALYST_PROJECT.

AnalystID	ProjectID	AnalystName	Dept	DeptCity	ProjectName	HoursLogged
A01	P10	Hamza Tariq	SOC	Lahore	Firewall Review	18
A01	P11	Hamza Tariq	SOC	Lahore	Threat Hunt	22
A02	P10	Nadia Mir	IR	Karachi	Firewall Review	30
A03	P12	Bilal Shah	SOC	Lahore	SIEM Config	12
A02	P12	Nadia Mir	IR	Karachi	SIEM Config	20

A). Answer each in one precise sentence.

i. AnalystName is the same in rows 1 and 2. Does this mean AnalystID alone can be the primary key of ANALYST_PROJECT? Why or why not?

ii. HoursLogged changes for the same analyst across different projects. What type of functional dependency does this tell you about HoursLogged relative to the primary/composite key?

iii. If we delete the only row for analyst A03, what information is permanently lost from the database — and what does this type of problem in unnormalised tables indicate?

iv. DeptCity is not directly determined by AnalystID alone — it is determined by Dept, which is determined by AnalystID. What is the name of this type of dependency, and which normal form eliminates it?

v. A student says: *"Since all cell values are atomic and there are no repeating groups, this table is fully normalised."* Identify the flaw in this reasoning in one sentence.

B). State whether ANALYST_PROJECT is in 1NF. Define 1NF and verify all its conditions against the given table. Identify the primary key.

C).

i. List all functional dependencies in ANALYST_PROJECT.

ii. Identify every partial dependency, naming which part of the composite key is the determinant and which attribute(s) depend on it.

iii. Decompose the table into 2NF relations. Name each table, list attributes, and underline the primary key.

D).

i. From your 2NF tables, identify any transitive dependency that remains. Write the full chain using $A \rightarrow B \rightarrow C$ notation.

ii. Decompose into 3NF. List all final tables with attributes and primary keys.

iii. State the total number of tables in your final 3NF schema.

E).

The last two rows are added to ANALYST_PROJECT.

AnalystID	ProjectID	AnalystName	Dept	DeptCity	ProjectName	HoursLogged
A01	P10	Hamza Tariq	SOC	Lahore	Firewall Review	18

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AnalystID	ProjectID	AnalystName	Dept	DeptCity	ProjectName	HoursLogged
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A02	P10	Nadia Mir	IR	Karachi	Firewall Review	30
A03	P12	Bilal Shah	SOC	Lahore	SIEM Config	12
A02	P12	Nadia Mir	IR	Islamabad	SIEM Config	20
A04	P99	Zara Noor	IR	Karachi	— (not yet assigned)	0

Answer each question to the point. Answers must name the anomaly type and its cause or fix — naming only the anomaly without to the point explanation receives 0 marks.

i. Analyst A02 appears with DeptCity = Karachi in row 3 and DeptCity = Islamabad in row 5. What type of anomaly is this, and which operation on the table causes it?

ii. If analyst A03's only row (P12) is deleted from the table, what information is permanently lost, and what is the name of this anomaly?

iii. Analyst A04 has been added with ProjectID P99, but no actual project details exist for P99. What type of anomaly does this represent, and what is the core reason it occurs in this table?

iv. Which specific violation in the table structure is the root cause of all three anomalies identified above — name the concept and the normal form that resolves it.

v. Suggest in one sentence how decomposing this table (as done in Parts 3 and 4) eliminates the update anomaly identified in part (i).

CLO 4

Question 3 **(20 +20 = 40 Marks)**

Table — ANALYST (use this table for all questions)

AnalystID	AnalystName	Dept	City	Salary	Certifications	HoursLogged
A01	Hamza Tariq	SOC	Lahore	75000	CEH	45
A02	Nadia Mir	IR	Karachi	55000	OSCP	38
A03	Bilal Shah	SOC	Lahore	62000	CISSP	52
A04	Aisha Noor	IR	Islamabad	48000	CEH	29
A05	Ali Raza	SOC	Karachi	70000	OSCP	41
A06	Amna Khan	Forensics	Lahore	58000	CISM	33
A07	Ahmed Butt	Forensics	Islamabad	61000	CEH	50

A). For each of the following scenario: name the relational algebra operation and write its expression.

i. The HR manager says: *"I only want to see the records of analysts who are working in the SOC department."* All columns of matching rows are needed.→ *Which operation is this? Write its expression using the ANALYST table.*

ii. The payroll team says: *"We need a list of analyst names and their salaries only — nothing else from the table."* All rows must be included, no row is filtered out.→ *Which operation is this? Write its expression using the ANALYST table.*

iii. A system report calls the result column Name but the table stores it as AnalystName. The data is the same — only the column label needs to change in the output.→ *Which operation handles this? Write its expression.*

iv. Query A returns AnalystIDs of all SOC analysts: {A01, A03, A05}.
Query B returns AnalystIDs of analysts with HoursLogged > 40: {A01, A03, A05, A07}.
The manager wants only the AnalystIDs that appear in both results.→ *Which operation is this? Write its expression using sets A and B.*

v. Set A = AnalystIDs of all Lahore analysts: {A01, A03, A06}.
Set B = AnalystIDs of analysts with Salary > 60000: {A01, A03, A05, A07}.
The manager wants Lahore analysts who do NOT earn above 60000 — only those in A but absent from B.→ *Which operation is this? Write its expression.*

vi. The firm acquires a second team whose records are stored in an identical table called ANALYST_B with the same columns. Management needs one combined list of all analysts from both tables with no duplicate rows.→ *Which operation is this? Write its expression.*

vii. An audit tool needs to generate every possible combination of one analyst paired with another analyst from the same ANALYST table — no conditions, just every possible pair of rows.→ *Which operation produces this? Write its expression. What will the size of the result be if the table has 7 rows?*

B). Write the SQL query for each scenario using the above ANALYST table. Maximum three lines per query.

i. The security lead checks analysts from the SOC department in Lahore every morning. He asks you to create a saved query so he never has to retype the filter — the result should show only AnalystName, Salary, and HoursLogged. *Create a view called SOC_Lahore that stores this query permanently.*

ii. Analysts are searched by Dept hundreds of times a day and the DBA notices these queries are slow on a large table. He wants to speed up Dept-based lookups without changing any data or rewriting any query. *Write the SQL to create an index called idx_dept on the Dept column of ANALYST.*

iii. The firm has a rule: no analyst's salary can be updated to below 40000. If someone tries to set a lower salary, the database must automatically block the update and raise an error message *'Salary too low'. Write a BEFORE UPDATE trigger called trg_salary that enforces this rule on the ANALYST table.*

iv. The operations manager wants to know which departments have more than one analyst who has logged more than 35 hours. He wants to see the department name and the count of such analysts.*Write the SQL using WHERE, GROUP BY, HAVING, and COUNT. Result must show only departments meeting the condition.*

v. Find the names of all analysts whose name starts with 'A' and who work in a department where at least one other analyst holds a CISSP certification.→ *Write the SQL using LIKE and EXISTS with a correlated subquery. Keep within three lines.*